PICASSA: Platform for Interactive, Comprehensive and Accessible Sc-RNA Seq Analysis - User Manual

Are you a wet-lab biologist struggling with single-cell RNA-seq data, and feeling overwhelmed by coding and endless back-and-forth emails with bioinformatics cores?

PICASSA is designed for you!

It empowers you to take full control of your data analysis, no programming required. From raw data to publication-ready results, PICASSA streamlines the process, making single-cell analysis interactive, comprehensive, and accessible.

PICASSA supports the analysis of single samples as well as multiple samples, including replicates.

Introduction

This manual provides a step-by-step guide for conducting single-cell RNA sequencing (scRNA-seq) analysis using PICASSA. Users should follow these guidelines to ensure accurate and efficient analysis. The application interface also provides concise guidance via the “Instructions” dropdown menu at each stage of the workflow. These in-app instructions provide a quick reference, while this manual offers more detailed explanations and is organized into subchapters for Phases 1 and 2, following the structure of the application interface to help users navigate the workflow more easily.

The application provides various customizable options, allowing users to adjust parameters such as colors, graph size, font size, and label orientation. All generated tables can be downloaded in Excel format, while plots in PNG and PDF format.

To ensure full traceability of the analysis workflow, users are strongly encouraged to download tables and figures as soon as they are generated.

PHASE1

Phase1 generates the initial UMAP and clustering of the data. This manual provides a step-by-step guide to help users complete the procedure.

To begin, click on “Open Phase 1” on the main interface.

Important Notes Before Starting

If the analysis has already been conducted up to the checkpoint at Step 5, you must load the seurat object (.rds file) saved at that step to proceed with the refined dataset for consistency of the analysis. Only the .rds file generated by PICASSA should be loaded, as files from other sources will not be compatible with the pipeline.

If you have not yet performed the analysis or wish to modify the filtering parameters, you should restart the analysis from the beginning.
Note: The Seurat object can only be saved once the workflow reaches Step 5 (Checkpoint). If the analysis is stopped before that step, the object cannot be saved, so it is recommended to progress at least up to Step 5 to preserve your current analysis state.

Based on your situation, choose how you would like to proceed under “A) Start with your data”:

- “Start a new analysis from scratch”, if you are beginning a new analysis or want to modify filtering parameters
- “Resume your previous analysis by loading your checkpoint.rds file (continue from Step 5)”, if your analysis has already reached Step 5

Alternatively, under “B) Start with a demo dataset”, you can explore PICASSA without using your own data. This section provides example datasets and includes options analogous to those in section A:

- “Load example data from the start (raw data)”, where experiment parameters and input files are automatically prefilled in the “Input Instructions” tab; click “Next” to proceed through the workflow step by step
- “Load example data from a Step 5 checkpoint”, where clicking “Run” loads data at the normalization step (Step 5), with filtering and normalization already completed; you can then continue through the rest of the workflow

This allows you to either run the full workflow or begin from an intermediate stage.

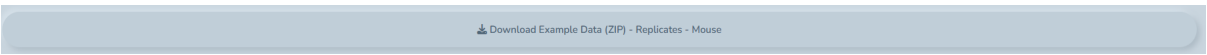
Continue the analysis from checkpoint

1. Load a previously generated seurat object (.rds file).
2. Select the organism used in the experiment.

Once the data is loaded and the "Run" button is clicked, the "5.Normalize - Dimensionality Reduction - Compute CCS" tab (5) will open automatically and the analysis continues from there. In the sections below, we will describe the workflow starting from the start through to the final steps. For clarity and consistency, the workflow described below follows the complete analysis from start to finish. The checkpoint functionality is introduced here and will be briefly referenced again at Step 5.

Start a new analysis from scratch

Input Instructions



1. Specify whether the experiment includes replicates or multiple samples in each group.
2. Select the organism used in the experiment.
3. Click “Let’s Start”.
4. Choose one of the data types below for the input files.
   Note: All samples must be provided in the same format:
   - Created by cellranger (most common).
     PICASSA automatically combines both Ensembl ID and gene symbol formats, allowing the dataset to include both.

        i. **10x – files:** barcodes.tsv.gz, features.tsv.gz, matrix.mtx.gz
        ii. **.H5 – file:** filtered_feature_bc_matrix.h5
    - Created by Scanpy or other AnnData based tools:
      The available gene identifiers (Ensembl IDs, gene symbols, or both) depend on how the original .h5ad file was generated.
        i. **.h5ad – file:** filtered_feature_bc_matrix.h5ad
    - Other alignment/counting tools:
      PICASSA attempts to detect and merge Ensembl ID and gene symbol columns automatically; if only one format is present, it will use that.
        i. **Gene-count matrix – file:** countMatrix.txt or countMatrix.tsv
5. Enter the sample name for which you intend to load the input files (i.e., Tumor).
6. "Browse" for the corresponding input file(s) and then "Load" it/them once the upload is completed.
   Note: The loading can take some time so be patient and wait until the "Load" button becomes clickable
   - Repeat steps 4-6 for each sample.
7. After loading the files for all samples, click "Done! Next".

Note: The input files are just temporarily uploaded in PICASSA server, once you exit the analysis they will be deleted.

8. Enter the group names in the desired order for downstream comparisons. The order you provide determines which comparisons will be performed in the downstream analysis (e.g., differential expression, pathway enrichment etc.). For example, to compare Tumor vs. Control, first enter the Tumor group and then the Control one.
9. Click "Next".

Note: If you are uploading replicates, each sample name must begin with its corresponding group name. Group names must be unique prefixes and should not be substrings of one another. Group names are case-insensitive, meaning that uppercase or lowercase letters do not matter; however, the group name must match the starting characters of the sample name.

Correct examples:

- Samples: bgl1, bgl2, pbs1, pbs2
  Groups: bgl, pbs
- Samples: CD1, CD2, HD1, HD2, RHDCD1, RHDCD2
  Groups: CD, HD, RHDCD

Incorrect examples:

- Samples: 1bgl, 2bgl, 1pbs, 2pbs
  Groups: bgl, pbs
  (The group name does not appear at the start of the sample name)
- Samples: HD1, HD2, HDCD1, HDCD2
  Groups: HD, HDCD
  (The group names are ambiguous because one is the prefix of the other)

1. Read Data + Cell Filtering

After clicking "Next", the "1. Read Data + Cell Filtering" panel will appear, containing a single "Run" button. At this step, PICASSA reads the input data, computes initial filtering thresholds, and generates several diagnostic plots.

- Customizable color assignments for groups and samples (if applicable).
- Two summary tables: one showing the number and percentage of cells per group or sample before and after filtering, and another comparing cell counts before and after applying each filtering parameter (e.g. UMIs, genes, mitochondrial and ribosomal percent). These tables update automatically when threshold values are changed, allowing users to preview the filtering impact before running the final step.
- Different Quality Control "(QC) plots before cell filtering" to help you examine population distributions and decide on thresholds for filtering cells appropriate to the specific experiment:
    - **Knee Plot:** The Barcode Rank Plot displays all barcodes before/after quality control, ordered from highest to lowest UMI count. It provides a visual summary of how filtering thresholds (e.g., minimum/maximum UMI counts, number of detected genes, and mitochondrial or ribosomal content) affect barcode retention.
    - **Cell Distribution Bar Plot:** It displays the number of cells per sample.
    - **Scatter Plot (Number of Genes vs Number of UMIs):** It shows the number of genes per cell vs number of UMIs per cell. It helps assess the relationship between these parameters, as well as any dispersion of data points. Cells appearing in the bottom right quadrant have a high number of UMIs but only a few numbers of genes, which may indicate dying cells or a population of low-complexity cell types (e.g., erythroid cells). Click here for more details.
    - **UMI Distribution Plot:** It shows the distribution of the number of Unique Molecular Identifiers (UMIs, corresponding to transcripts) detected in each cell. Markedly higher UMI counts relative to the main distribution may indicate multiplets (i.e., a droplet in which multiple cells were captured during the microfluidic encapsulation), whereas markedly lower UMI counts might be droplets containing low-quality (dying, degraded or damaged) cells or ambient RNAs. It is important to keep in mind that different cell types and cells isolated from various species may differ in their total RNA content. For example, tumor samples often contain large tumor cells, which typically have higher RNA content, mixed with smaller tumor-infiltrating lymphocytes, which generally have lower RNA content. **Note:** The minimum UMI threshold is automatically pre-filled based on Cell Ranger's default cutoff, where barcodes with fewer than 500 UMIs are not considered real cells. Before proceeding with the analysis, it is recommended to inspect the barcode rank plot. If a warning appears or the plot lacks a clear "steep cliff" pattern, the dataset may contain a high level of ambient RNA, and additional quality control such as ambient RNA removal may be necessary (check here for more details).
    - **Gene Distribution Plot:** It shows the distribution of the number of unique genes expressed per cell. Similar to the observation with UMI counts, barcodes associated with unusually high numbers of genes might be multiplets, whereas barcodes with low number of genes might be droplets containing low-quality cells or ambient RNAs. However, it should be noted that biological differences between cell types and species can affect this metric. For example, stem cells may express a higher number of unique genes. **Note:** The minimum number of genes threshold is automatically pre-filled based on.

- **Mitochondrial Percentage Distribution Plot:** It shows the distribution of the percentage of UMIs mapping to the mitochondrial genome per cell. Stressed and damaged cells often exhibit high mitochondrial content, possibly because of increased apoptosis (during which mitochondrial membranes lose integrity and mitochondrial RNA is released into the cytosol) and/or loss and degradation of cytoplasmic RNA from lysed cells and concomitant apparent enrichment of mitochondrial RNA that is protected by double membranes and is therefore among the last to degrade. This paper suggests that a threshold of 5% is optimal for mouse data and 10% for human data. However, it is important to take into consideration that mitochondrial transcript abundance varies with cell type and species. For example, metabolically active cells may express a higher number of mitochondrial genes, and filtering based on mitochondrial reads could introduce bias in the analysis.
- **Ribosomal Percentage Distribution Plot:** It shows the distribution of the percentage of UMIs mapping to the ribosomal genome per cell.
- **Violin Plots:** Alternative visualization of plots 2-5.
  Note: You can click "Download all plots" to export a PDF with the "QC plots before cell filtering". If replicates or multiple samples exist, all of the above plots will also be generated per sample, selectable from the dropdown menu.

- Customizable thresholds for the following cell-filtering parameters:
  - **UMIs:** Number UMIs detected in each cell. **Note:** The minimum value calculated by CellRanger for calling something a "cell" is set as the default minimum value. In addition, dataset-specific suggested limits are computed using the distribution of detected UMIs: the median is used to define an adaptive range (median ± 5 × SD). This dynamic interval provides guidance for refining lower and upper UMI cutoffs according to the dataset's variability. See this link for more details.
  - **Genes:** Number of unique genes detected in each cell. **Note:** Similar to the observation above, dataset-specific suggested limits are computed using the distribution of detected genes: the median is used to define an adaptive range (median ± 5 × SD). This dynamic interval provides guidance for refining lower and upper UMI cutoffs according to the dataset's variability. See here.
  - **Mitochondrial Percent:** Percentage of UMIs that map to the mitochondrial genome per cell. **Note:** The maximum value is set by default to 5 for mouse data and 10 for human data as suggested by this paper.
  - **Ribosomal Percent:** Percentage of UMIs that map to the ribosomal genome per cell.
- **Apply thresholds for cell filtering:**
  - Set desired threshold values for the relevant cell-filtering parameter(s). If no value is provided for a threshold (left blank), the system will automatically assign the respective minimum or maximum value for that parameter. **Note:** Thresholds are shown in the "QC plots before cell filtering" as red lines for clear visualization of the cutoff point in the population. When threshold values are changed, the red lines update automatically.
  - Click "Run" to apply the selected thresholds for the different cell-filtering parameters and obtain "QC plots after cell filtering" accessible in the subpanel "QC plots after cell filtering". You will be automatically redirected to the "QC

plots after filtering" subpanel. **Note:** If, after running the cell-filtering, a change in thresholds for one or more parameters is desired, return to the "QC plot before cell filtering" subpanel, set new values, and click the "Run" button. The plots in the "QC plots after filtering" will then be updated accordingly. You can click "Download all plots" to export a PDF with the "QC plots after cell filtering".

- Click "Next Step" to access the next step of the analysis.

2. Doublet Detection - DoubletFinder

Doublets occur when two cells are mistakenly encapsulated in one droplet, leading to misleading results. This paper compared different computational doublet detection methods for scRNA data, and DoubletFinder was identified as the method with the best detection accuracy among the evaluated methods. High-confidence doublets typically represent heterotypic doublets, where two different cell types are captured together. Low-confidence doublets may represent homotypic doublets, which occur when cells of the same type are captured together and are therefore more difficult to distinguish from singlets. If predicted doublets form small, distinct clusters in the embedding, their removal is generally recommended.

This step is optional and can be skipped by clicking the "Skip" button. This option is provided for users who wish to quickly explore the dataset, as DoubletFinder can be computationally intensive and may require additional processing time. If one wishes to proceed then:

1. Click "Run" to generate a Uniform Manifold Approximation and Projection (UMAP) plot displaying predicted single cells (singlet) and doublets, including both high-confidence (heterotypic) and low-confidence doublets (potentially homotypic), as determined by DoubletFinder. The DoubletFinder github page explains this in more depth. **Note:** If doublets create small clusters, it is especially recommended to remove them.
2. Before making this decision, click "2.Check Metrics" to review the detailed counts of singlets and predicted doublets.
3. Select "Only High Confidence" or "All" under the "Remove Doublets or Not" option to visualize how each doublet-removal strategy affects the UMAP and to decide which doublets should be excluded. If you do not wish to remove any doublets, select "Do not remove". To proceed, choose one of the three options and click "Run". Once the process is complete, the "Next Step" button will be enabled, allowing you to continue with the analysis. After completion, download the updated plots and proceed by clicking "Next Step." To assist in selecting the appropriate option before clicking "Run", plots are provided to help identify potential doublets in the dataset:

The following plots illustrate how predicted doublets are distributed in the dataset:

- **Overview:** Cell embedding colored by DoubletFinder classification (Singlet, Doublet – Low Confidence, Doublet – High Confidence).
- **Group-level:** The same embedding split by experimental Group.
- **Group-Violin:** The same embedding split by Sample, if sample information is available.
- **Sample-level:** Distribution of QC metrics per Group.
- **Sample-Violin:** Distribution of QC metrics per Sample, if available.

Each violin is splitted according to the DoubletFinder classification to allow comparison between singlets and predicted doublets.
The displayed metrics of the violin plots include:

- ➔ UMI counts (*nCount_RNA*)
- ➔ Gene counts (*nFeature_RNA*)
- ➔ Percentage of mitochondrial genes
- ➔ Percentage of ribosomal genes

Note: If doublets do not form distinct small clusters and you are uncertain about removing them, it is recommended to proceed without filtering at this stage. Following downstream analysis, you can revisit this decision and determine whether doublet removal is appropriate. Color customization options are also available to distinguish between different doublet types.

4. Download the table present in the “Metrics Data Summary” subpanel. This table contains useful information on the quality of the cells and genes that were left after cell filtering, and, if applied, after doublet removal, both per sample and across the entire dataset (all cells combined, indicated as “Total”).

Metrics Data Summary

Independently of running doublet detection, the Metrics Data Summary provides additional information that can assist in determining appropriate thresholds for Step 3: Filter Genes. Thresholds in the next step:

- **Include genes detected in at least this many cells:** Input the minimum number of cells that express the same gene. The 'Percentage of genes expressed in N or fewer cells (including genes with zero counts in all cells)' and 'Percentage of genes expressed in N or fewer cells (excluding genes with zero counts in all cells)' rows helps in defining this number. For example if a threshold of 3 is used then this will output genes expressed in 3 and more cells.
- **Get genes that express at least this number of counts across cells:** Input the minimum number of counts for each gene. The 'Q0 of counts per gene' gives the minimum number of counts of a gene and the 'Percentage of genes with total counts of N or fewer across all cells' rows, also helps define this number.

Note: Running the doublet detection and UMAP generation multiple times with the same parameters will produce the same results, as PICASSA fixes the random seed internally for reproducibility. However, re-running the analysis after changing upstream parameters (e.g., cell filtering thresholds) may yield a different UMAP layout. This occurs because UMAP is a nonlinear dimensionality reduction algorithm whose structure depends on the specific set of input cells and features, modifying these inputs alters the underlying neighborhood graph, leading to slight shifts or rearrangements in the embedding.

3. Filter Genes

Users must define filtering thresholds based on the Metrics Data Summary table as explained prior:

1. **Minimum number of cells per gene:** Determines how many cells must express a gene for it to be retained.
2. **Minimum expression counts per gene:** Ensures genes with low expression are filtered out to avoid artifacts.

These thresholds help refine the dataset, reducing technical noise while preserving biologically relevant genes. This panel generates the same QC plots as in step *"1. Read Data + Cell Filtering"* from *"Start a new analysis from scratch"* and a *"Metrics Data Summary"* table.

4. Cells Summary

If DoubletFinder was performed, two tables are generated. In both tables, rows correspond to groups/samples, and columns provide information about cell counts and proportions. The first table contains the full set of metrics prior to doublet removal, while the second summarizes the data after applying the selected doublet-filtering strategy. Percentages for "Cells After Filtering" are calculated relative to the total number of "Cells Before Filtering". All remaining percentages are calculated relative to the total number of "Cells After Filtering". If DoubletFinder is not performed, only a single table is generated, without annotations for singlets and doublets. ***Please click "Download Report" to download an Excel file summarizing the parameters selected up to this step for future purposes and to activate the Next Step button allowing you to proceed.***

5. Normalize - Dimensionality Reduction - Compute CCS

There are no user-defined parameters at this step. The following section provides background information to help interpret the results:

Principal Components (PCs) Overview

Principal Components (PCs) represent a set of genes that summarize most of the dataset's variance. Selecting an appropriate number of PCs is essential:

- If too few PCs are chosen, critical data may be lost.
- If too many PCs are selected, noise may overshadow meaningful data.

The preferred number of PCs is automatically calculated and suggested; however, users have the option to modify this parameter in the subsequent step.

Calculation of Preferred PCs

The selection of the number of PCs is determined based on the minimum value of two criteria:

- The cumulative sum of the standard deviation percentage must be greater than 90%, while the individual standard deviation percentage should be less than 5%.
- The last PC where the percentage variation changed from the previous one exceeds 0.1%.

Plots generated:

- **Elbow Plot:** Displays the PCs (x-axis) versus their respective standard deviation (y-axis). A vertical line marks the recommended number of PCs. This plot helps the user to define the number of PCs.
- **Dot Plots:** Highlight the most highly variable genes, specifically those that exhibit significant cell-to-cell variation.
- **PC Loadings List:** Shows the top genes with the most positive and negative loadings for the first two PCs, representing gene modules that correlate or anti-correlate across single cells. Loadings indicate how much each gene contributes to a given PC.

- **PC Heatmaps:** Displays heatmaps for the first preferred number of PCs + 5. Greater contrast indicates more data explained by the gene, helping identify where noise appears. These plots help the user to define the number of PCs.
- **Correlation Plot:** Shows the contribution of each variable to different PCs. The larger and darker a dot, the higher its contribution. Colours (blue/red) indicate correlation (positive/negative), but the key takeaway is the dot's intensity rather than its specific colour. This plot helps determine which variables significantly contribute to PCs. If cell cycle genes dominate early PCs, they might influence clustering and marker gene identification.
- **PCA on Cell Cycle Genes:** If the colours in the PCA plot separate distinctively, it indicates that cell cycle genes significantly impact the dataset.
- **Grouped Cell Cycle Phase Plot:** Similar to the previous PCA plot but grouped by cell cycle phase, allowing a clearer understanding of its influence on clustering and downstream analysis.

Depending on the experiment, users may choose to regress out the cell cycle phase effect if it interferes with clustering. However, in some cases, preserving this effect is beneficial and should not be removed. Hence it is advised to proceed without any regression (if the experiment is also fitted to do so) and continue analyzing the data. When in doubt, the user can continue from the checkpoint and choose to regress variables out and compare results.

Please click "Download seurat object" to download an .rds file containing all analyses completed so far. This file acts as a checkpoint for reloading the data and activates the "Next Step" button to continue.

6. Regression

If "Yes" is selected, users can regress multiple variables to remove unwanted sources of variation such as cell cycle phase, otherwise, the analysis proceeds to the next step automatically.

For regressing out the cell cycle phase, users can choose:

- **S.Score and G2M.Score:** These scores indicate cell cycle activity. OR
- **CC.Difference:** This variable accounts for both S.Score and G2M.Score, providing a combined effect.

  Note: If you choose to regress, please choose either S.Score and/or G2M.Score OR CC.Difference. PICASSA will give a warning if the wrong input is given.

More details on cell cycle regression are available in this link.

The same set of plots from the previous step is generated again to confirm the effectiveness of the regression process and to determine how the preferred number of PCs changes post-regression.

7. Determine PCs

Users have the option to modify the number of PCs, which is automatically pre-filled based on prior calculations. To reduce the technical noise in single-gene expression for scRNA-seq data, the Seurat package groups cells into clusters based on their PCA scores. These scores come from the expression of the most variable genes, where each PCs represents a 'metagene' that summarizes information from related genes. Selecting an appropriate number of PCs is critical for capturing the majority of meaningful variation and avoiding unnecessary noise that could obscure biological signals.

At this step, UMAP visualizations are generated coloured by groups/samples. These plots assist in deciding whether to proceed with data integration or merging in the next step.

- **Integrate:** If groups or samples appear clearly separated, integration is typically recommended. Integrative analysis aligns shared cell types and states across datasets, improving comparability and increasing statistical power. It also corrects for batch effects arising from factors such as different tissues, sequencing runs, time points, or technologies.
- **Merge:** When the groups or samples are uniformly distributed usually merge is chosen. There are no batch effects.

Choosing Between Integration and Merging In Summary:

- **Integration:** Recommended when groups or replicates are highly distinct and separate. This approach corrects batch effects caused by factors such as:
    - Different tissue sources.
    - Sequencing performed at different times.
    - Variations in sequencing technology.
- **Merging:** Preferred when groups or replicates are uniformly distributed, suggesting no significant batch effects exist.

By determining whether integration or merging is appropriate, users can ensure optimal clustering and downstream analysis.

8. Multiple plots/CCS

The first step, based on the previous step, is to decide whether to merge or integrate the datasets. If integration is selected, a list of available integration methods will be displayed. This step can be performed multiple times using different methods, as there is no universally optimal approach; the most suitable method depends on the characteristics of the specific dataset. However, integration can be computationally intensive and should be approached with care. It is recommended to begin with the default method (Harmony) and only explore alternative approaches, such as CCA or RPCA, if the overlap between conditions is not satisfactory. Repeatedly running integration is discouraged, as it can consume substantial memory resources and may lead to session instability or crashes. Below is a summary of the available integration methods:

Integration Methods

Users can select different methods for batch correction:

- **CCA (Canonical Correlation Analysis):** Captures shared biological variation across datasets but can be computationally intensive for large datasets.
- **RPCA (Reciprocal PCA):** Faster and more memory-efficient for large datasets, though it may perform less well when batch effects are very strong.
- **Harmony:** A scalable method that effectively corrects strong batch effects but may require parameter tuning to avoid overcorrection.

Note: If integration does not result in meaningful changes in group or sample distribution, selecting the **merge** option may be more appropriate.

Then the user needs to define dimensionality reduction options:

- **K parameter:** Defines the number of neighboring cells considered during clustering. The default value is 20. Multiple values can be tested by providing a comma-separated list without spaces (for example: 10,20).
- **Resolution range:** Controls cluster granularity, with higher values producing more clusters. The accepted range is 0.1 to 2, with commonly used values between 0.1 and 1. Providing both a starting and an ending value allows testing a range of resolutions.

Multiple plots are generated based on clustering results and cell cycle effects. Clustering visualizations guide the selection of final clustering parameters, while cell cycle plots support the assessment of cell cycle phase, which may be important in certain analyses.

Clustering

- **Dimensionality reduction plot:** Coloured by groups to visualize clustering.
- **Clustering Trees:** Generated for each K parameter value, illustrating relationships between clusters at different resolutions.
    - **Nodes (Circles):** Each node represents a cluster at a given resolution, sized according to the number of data points (cells) in that cluster.
    - **Edges (Lines):** Represent transitions between clusters across resolutions. Brighter and more vibrant edges indicate a higher number of transitioning data points.
    - **Cluster Assignments at Different Resolutions:** Displays clustering at multiple resolution levels, helping to refine selections.
- **Multiple dimensionality reduction plots:** Generated based on selected parameters. These plots allow for an iterative approach, enabling further investigation into clusters before making a final decision in Phase 2.

Cell Cycle Effect

- **Dimensionality reduction plot:** Coloured by groups to visualize clustering.
- **Dimensionality reduction plot:** Coloured by cell cycle phase to observe cell cycle distribution.
- **Dimensionality reduction plot:** Split by cell cycle phase and coloured by groups, allowing a combined view.
- **Dimensionality reduction plot:** Showing distinct cell cycle phases across the dataset.
- **Bar plot:** Displaying the percentage of cells in each cell cycle phase per group.

9. Final plots/Marker genes

Final UMAP visualization is generated using the selected parameters, including the optimal K and resolution values. Marker genes are then identified by comparing each cluster against all others, based on user-defined expression thresholds.

Expression parameters for marker genes:

- **Average Log2 Fold Change threshold:** A value of log2FC = |1| means that the expression change is double or half (FC = 0.5 or 2). The FC can be found using 2 to the power of the log2FC value, e.g.,
  If log2FC = |1| then FC = 2^1 = 2 and 2^{-1} = 0.5.

- **Minimum percentage threshold:** The gene is at least, e.g., 25% expressed in the first or second group of comparison.
- **Minimum difference of percentages threshold:** The gene has at least, e.g., 25% difference of expression between the first and second group of comparison.
- **Adjusted P-value Threshold**: Controls statistical significance after correcting for multiple hypothesis testing. Differential expression is assessed using the Wilcoxon rank-sum test (test.use = "wilcox"). In Seurat's FindMarkers() function, p-values are adjusted using the Bonferroni correction by default, which is a conservative method that reduces false positives by accounting for the total number of genes tested. Genes with an adjusted p-value below the selected threshold are considered statistically significant.

<u>In this step the output is shown in multiple different subpanels explained below:</u>

➔ **Save some data:**

Save objects

- Download seurat object: Download the final seurat object containing all processed data, metadata, and analysis results.

Save distribution tables

- Save Distribution Tables: Download summary tables showing the distribution of cells across clusters, groups, or samples.

Genes Between Clusters

- Save Marker Genes - no thresholds: Download all marker genes identified between clusters without applying filtering thresholds.
- Save Marker Genes - with thresholds: Download marker genes between clusters after applying the selected filtering thresholds.
- Save GSEA prerank - Clusters: Download preranked gene lists for Gene Set Enrichment Analysis (GSEA) based on cluster-level comparisons.

Next Step

- Save Seurat object to activate the next step: Saving the final Seurat object enables the Next Step button, allowing you to proceed to downstream analyses.

➔ **Final plots**

Choice to change colours for the down- and up-regulated genes.

Main Plots

- Elbow plot showing, with a vertical line, how many PCs were selected.
- Final UMAP plot.
- Bar plot showing the distribution of up- and down-regulated top marker genes in each cluster.
- UMAP plot colored by cell cycle phase.

Group-level Plots

- UMAP plot split by cell cycle phase and colored by groups/conditions.
- Bar plot showing the distribution of cell cycle phases across clusters.
- UMAP plot split by groups/conditions, showing the final clusters.
- UMAP plot split by groups/conditions.
- Bar plot showing the distribution of cells from each condition in every cluster.

Sample-level Plots (if replicates are present)

- UMAP plots per sample colored by cell cycle phase.
- UMAP plots per sample showing clusters.
- UMAP plots per sample split by clusters.
- Bar plot showing the distribution of cells per sample.
- Bar plot showing normalized cluster distribution per sample.

➔ **Cluster Quality Control (QC)**

- **Feature and violin plots** displaying: Number of genes and UMIs after filtering (log10 scale). Percent of mitochondrial and ribosomal counts.

➔ **Top 10 Marker Genes**

- **Table** displaying the top 10 marker genes in each cluster, ordered by: a) Average Log2 Fold Change (log2FC) value and then b) P-adjusted value

➔ **Top 5 Marker Genes Feature Plots**

- **Feature plots** showing the expression of the top 5 marker genes in the chosen cluster. Options available: Choose the colour for feature plots. Select which cluster to display.

➔ **Top 5 Marker Genes Violin Plots**

- **Violin plots** showing the expression of the top 5 marker genes in the chosen cluster. Options available: Select which cluster to display.

➔ **Top 5 Marker Genes Dot Plots**

- **Dot plots** showing the average expression of the top 5 marker genes in the chosen cluster. Options available: Choose the color for dot plots. Select which cluster to display.

10. Report

Download the report to save all steps performed.

11.Check

Unsaved items are listed in the table below. After saving, click "Check" again to refresh the table, then completed items will no longer appear.

PHASE2

Phase2 provides a more interactive environment and does not necessarily follow a strict step-by-step workflow. It includes a wide range of downstream analyses such as cell type annotation, differential expression analysis (DEA), trajectory inference, gene expression exploration, enrichment analysis, GSEA, and more. Below is a brief overview to guide the user through the required inputs and expected outputs for each analysis.

Initial Considerations

To begin, a Seurat .rds object must be loaded. The user is then asked to specify the organism used and indicate whether the object was generated within PICASSA Phase 1. If the object was not created in Phase 1, a conversion step is performed to ensure

compatibility with the platform. During this process, the user defines key metadata columns, such as cluster identities, group labels, and optionally cell cycle scores, among others. These inputs are straightforward to provide and enable seamless integration of external datasets into the PICASSA workflow.

Clustering

Once the data are successfully loaded, options to specify colours for volcano plots, groups/conditions, and samples/replicates (if applicable) will appear. These colours will be used throughout all subsequent analyses, so it is recommended to define them at this stage. The respective dimensionality reduction plot is shown along with the number of cells in every cluster. Here there is the option to modify clusters, more specifically, to merge or rename clusters, to divide, and to remove. If removal is performed, then a new button appears that leads the user to perform some prefilterings steps (also included in Phase1), to clean the data, renormalize and create a new final dimensionality reduction plot. This is the only time in Phase2 that the user needs to follow steps, otherwise, the analysis can be run independently of each other.

Final results

This module mirrors the functionality of *Phase 1 – "9. Final Plots / Marker Genes"*. It is provided to allow users to regenerate final outputs after modifying the clustering structure in Phase 2 or simply to reproduce results from Phase 1, ensuring that results reflect the updated dataset when applicable. As such, this module represents the final stage of preprocessing, enabling users to obtain the updated clustering alongside relevant summary information and marker gene results.

Expression Analysis and Cell Enrichment

This panel begins with the option to modify clusters using the same functionality provided in the "*Clustering"* panel. This enables more refined exploration of the data, supporting improved cluster annotation, identification of potential contaminants, and more accurate interpretation of results. Once the desired cluster modifications have been defined, three subpanels become available for further analysis.

Gene Expression

In this module the user specifies the genes of interest, either by manually providing gene names or by selecting predefined GO pathways. To facilitate clear labeling and organization of the resulting plots, the user is also required to assign a custom name to the selected gene set. The output of this module is a table along with some plots to help the user see results clearer.

- The **table** shows the total number of input genes, the number of genes found in the dataset and the genes not detected in the dataset. This helps assess coverage and ensures correct gene matching.
- Then a **feature plot** is shown which visualizes spatial expression of selected genes on the embedding, with customizable color gradients and option to select individual genes for display.
- **Violin plots** allow comparisons of expression patterns across different contexts: *Per Condition* (comparing groups such as treatments), *Conditions within Clusters*

(highlighting condition-specific differences within each cluster), and *Per Cluster* (showing expression distributions across clusters).

- In addition, a **dot plot** summarizes gene expression across clusters, where dot size represents the percentage of expressing cells and color indicates the average expression level.
- Finally, a **heatmap** visualizes expression patterns across selected clusters and/or conditions, enabling a more global overview of gene behavior. All plots are highly interactive and customizable, allowing users to adjust parameters such as font sizes, axis angles, colors, and plot dimensions to optimize visualization and interpretation.

Average Expression

This module enables users to explore the average expression of selected gene sets across clusters and conditions, with the flexibility to modify the clustering structure prior to visualization. After finalizing cluster selection, the user defines the gene sets of interest using one of the following approaches: (i) manually selecting genes, (ii) choosing predefined GO pathways, or (iii) uploading a file containing known marker genes. For the upload option, the file must be in .xlsx format, where each column represents a gene set or cell type and contains genes in Ensembl ID format. To facilitate the required format, PICASSA provides a helper app called *"Gene Conversion"*, which enables users to convert gene names or orthologs (e.g., human or mouse) into the appropriate Ensembl IDs.

- A summary **table**, similar to the one in the Gene Expression module, is generated to display the total number of genes loaded, the number of genes found in the dataset, and those not detected. This helps assess coverage and ensures correct gene matching.
- **Feature** plots display the spatial distribution of the average expression of each gene set on the embedding, with customizable color gradients and selectable gene sets or cell types.
- **Violin** plots show the distribution of average expression values across clusters or conditions, with options to adjust axis labels, angles, and perform statistical comparisons using Mann U Whitney test.
- In addition, **dot** plots summarize average expression across clusters, where dot size reflects the proportion of expressing cells and color represents the average expression level.
- Furthermore, **AUCell** plots are available to evaluate gene set activity at the single-cell level by calculating enrichment scores (AUC) for each gene set, providing an additional layer of interpretation.

All plots are highly interactive and customizable, allowing users to adjust parameters such as font sizes, axis angles, colors, and plot dimensions to optimize visualization and interpretation.

Single Cell Enrichment Analysis

This step computes **gene set enrichment scores at the single-cell level**, allowing the activity of biological pathways or functional gene sets to be evaluated across individual cells and cell clusters. The analysis can be repeated multiple times using different pathway databases, gene sets, or enrichment methods. Then the user selects the database or source used to define the gene sets for enrichment analysis. Available options include:

- Gene Ontology (GO) pathways: Gene Ontology provides curated functional annotations describing biological roles of genes. Users can select specific GO pathways to evaluate their activity across cells.
- MSigDB pathways: MSigDB contains curated collections of gene sets derived from biological pathways, regulatory signatures, and experimental datasets. Users can select pathways from the MSigDB collections.
- Upload a custom gene set file. Users may upload an Excel file containing custom gene sets. This option allows enrichment analysis using user-defined pathways or gene lists.

Note: After selecting a database (GO or MSigDB), the available pathways will appear. Users may select **one or multiple pathways** for enrichment analysis. Only genes that are present in the dataset will be used in the analysis.

Finally the user selects the enrichment method which is an algorithm used to calculate enrichment scores for each cell. Available methods include:

- UCell: Computes enrichment scores using a **rank-based Mann-Whitney U statistic**. Genes are ranked per cell, and the enrichment score reflects whether genes from the selected gene set are consistently ranked higher than background genes. UCell is optimized for sparse single-cell datasets and is typically the **fastest method**.
- AUCell: Evaluates whether genes from a gene set are enriched among the **top-ranked genes in each cell** by calculating the **Area Under the Curve (AUC)**. This method is widely used in single-cell regulatory network analysis.
- ssGSEA Single-sample Gene Set Enrichment Analysis: calculates enrichment scores based on **cumulative distribution functions of ranked gene expression values**. This approach is widely used but may be slower for very large single-cell datasets.

Each cell receives an enrichment score for each selected gene set. These scores can be visualized through several output tabs, similar to “Gene Expression” and “Average Expression” panels.

- A **data info table** is produced summarizing how the selected gene sets map to the genes present in the dataset. For each gene set, the table reports the name of the pathway or gene set, the total number of genes originally defined in the pathway, the number of genes from the pathway that are present in the dataset and finally the gene names from the pathway that were not detected in the dataset. This information helps evaluate the coverage of each pathway within the dataset.
- **Feature, violin and heatmaps** are also generated to display the enrichment score of a selected gene set across the whole dataset.
- **Heatmaps** summarize the average enrichment score of each gene set across clusters. Rows represent gene sets, while columns represent clusters. This visualization allows users to compare pathway activity patterns between gene sets across cell populations. Heatmaps provide a global overview of pathway activity across the dataset.

Annotation Tools

The tool and its respective tissue needs to be defined. The tools used are scType and scMayoMap. For scMayoMap to be used the marker genes for each cluster are needed so “Final UMAPs” need to be run to produce them. Below is a brief description of each annotation method:

- **ScType**
  ScType is an ultra-fast, fully automated tool for identifying cell types from single-cell RNA-seq (scRNA-seq) data. It offers a complete pipeline that includes data processing, normalization, clustering, and cell type annotation using a comprehensive marker gene database.
- **scMayoMap**
  scMayoMap is designed for cell type annotation in scRNA-seq datasets using a rich marker database. It includes information on 340 cell types across 28 tissue types, enabling accurate and user-friendly annotation.
  Note: This method will only be available if the *"Marker Genes"* step under *"Final UMAPs"* has been executed.

For every annotation method, a results table with cell type annotation scores and a set of plots for visualization is produced. Regardless of the annotation method used, each run will generate a dimensionality reduction plot showing the predicted annotated clusters. This provides a visual overview of how cells have been grouped and labelled by the method. Each method also produces a second, method-specific plot to further support interpretation of the results:

ScType

- **Barplot of Annotation Scores**: Displays the annotation scores for each cluster where x-axis are the cell clusters, y-axis are the annotation scores and above each bar, a label of “Yes” or “No” which indicates whether the cluster passed the annotation threshold.

scMayoMap

- **Dot Plot of Annotated Cell Types**: Visualizes the expression of marker genes across clusters, helping confirm and interpret the assigned cell types.
  The dot size and colour typically reflect expression levels and the proportion of expressing cells, respectively.

Projections

Projection analysis allows you to load a reference Seurat object, representing a known single-cell RNA-seq experiment using its .rds file. This reference can help interpret and define clusters in your current dataset, especially if the experiments are similar.
To perform a projection analysis:

1. **Load Reference Seurat Object**
   Select a Seurat object in .rds format to load and use for projecting clusters.
2. **Set Log2 Fold Change Thresholds**
   Define the Log2FC thresholds for identifying differentially expressed genes (DEGs) in both the current and reference datasets.
3. **Set Minimum Percentage Thresholds**
   Specify the minimum percentage of cells expressing a gene in each group, used to detect DEGs.
4. **Set Adjusted P-Value Thresholds**
   Define the adjusted p-value thresholds to determine statistical significance for DEGs in both datasets.

Once the projection is complete, a series of visual outputs are generated to help interpret how the datasets relate to one another. The outputs depend on the direction of the projection: The first two are from the same method and the last one is from the method that Seurat uses.

- **For the first one**, basically the goal is to map clusters from one seurat object (reference or data) onto another (data or reference) by comparing their differential gene expression profiles, quantify the similarity and produce visual summaries. For each cluster pair, it finds overlapping marker genes and computes two scores: a) Set Overlap Score (S): based on proportion of shared genes and b) Expression Score (E): based on shared expression magnitudes. Taking these 2 scores, a final score is then produced, by simply multiplying them together.
- The Seurat method **projects reference data onto a query dataset**, it uses a process called **reference mapping**, which is designed to transfer cell type labels, embeddings, or other annotations from a well-annotated reference to a new, possibly unlabelled query dataset.

When Your Data is Projected onto the Reference Data

- **Table**
  The table summarizes pairwise comparisons between clusters from your data and the reference dataset (represented by the *cluster_1* and *cluster_2* columns, respectively), quantifying the overlap between their marker genes. For each cluster pair, it reports overlap-based metrics (S score, E score, and the combined Overlap Score), along with the number of shared marker genes and the total number of marker genes in the reference cluster.
- **Alluvial Plot**
  Displays the top five genes per cluster. Highlighted lines represent the genes with the maximum log2 fold change (Log2FC) for each cluster compared to the reference.
- **Violin Plot**
  A violin plot shows the average expression of genes from your data's clusters, projected onto the clusters of the reference dataset shown on the x-axis.
- **Feature Plot**
  A feature plot displays the expression of genes from your data's clusters overlaid on the reference dataset's UMAP or spatial layout.

When the Reference Data is Projected onto Your Data

- **Table**
  The table summarizes pairwise comparisons between clusters from the reference dataset and your data (represented by the *cluster_1* and *cluster_2* columns, respectively), quantifying the overlap between their marker genes. For each cluster pair, it reports overlap-based metrics (S score, E score, and the combined Overlap Score), along with the number of shared marker genes and the total number of marker genes in the reference cluster.
- **Alluvial Plot**
  Focuses on the top five genes for each cluster, with highlighted lines indicating those with the highest Log2FC when projected onto your data.
- **Violin Plot**

A violin plot showing the average expression of genes from the reference clusters, projected onto the clusters in your dataset shown on the x-axis.

- **Feature Plot**
  A feature plot visualizing the expression of genes from the reference clusters, mapped onto your data's UMAP or spatial representation.

When Seurat method is used

- A table showing a summary of the prediction scores transferred from the reference to the query dataset, these are the scores Seurat assigns when it maps a cell to a specific identity.
- Generates a UMAP plot of the query dataset, coloured by the transferred annotations.
- Creates a bar plot showing the mean prediction score for each possible label.
- Creates a violin plot of prediction scores across all cells for each label.
- Generates a bar plot showing the number of cells assigned to each predicted identity.

Differential Expression Analysis (DEA)

This module allows comparison of gene expression, **between conditions across all data, between conditions within each cluster**, **between specific clusters and between specific conditions from a specific cluster**. In all cases, the analysis returns a table of differentially expressed genes (DEGs), along with summary plots such as volcano and violin with boxplot overlay.

Between-Conditions across all data

This section performs differential expression analysis between user-defined conditions across the entire dataset. Although similar comparisons can be generated in the "Final Results" tab, those analyses are based on the initial selections provided earlier in the workflow. In contrast, this section allows users to define custom comparisons, providing greater flexibility for exploring condition-specific differences.

- **Select the number of comparisons:**
  Users first specify the number of comparisons they wish to perform. Each comparison represents a differential expression test between selected conditions.
  Examples:
  - Condition 1 vs Condition 2
  - Condition 1 vs Condition 2 + Condition 3

  In this case, the total number of comparisons would be two.
- **Define Comparison Groups**
  For each comparison, users must define the conditions that will be included in the two groups being compared. This allows flexible grouping of experimental conditions when testing for differential expression.

Between-Condition Comparison Within Each Cluster

This analysis compares different conditions (e.g., treatment groups) **within each selected cluster**.

- **Select Clusters**
  Choose the clusters in which you want to perform the condition comparisons.

- **Select Condition Comparisons**
  Define which conditions to compare within each cluster.

Between-Specific Clusters
This analysis compares different clusters.

- **Specify the Number of Comparisons**
  Enter the total number of comparisons you want to perform.
  Example:
    - Cluster 1 vs. Cluster 2
    - Cluster 1 and 3 vs. Cluster 2

  In this case, the number of comparisons is **2**.
- **Define Clusters to Compare**
  Manually input the clusters involved in each comparison.

Between-Specific conditions from a specific cluster

- **Specify the Number of Comparisons**Enter how many unique comparisons you want to perform.
  Example:
    - Cluster 1, Condition 1 vs. Cluster 2, Condition 1
    - Cluster 1, Condition 1 vs. Cluster 2, Condition 2 and Cluster 3, Condition 2

  In this case, the number of comparisons is **2**.
- **Define the Comparisons**
  Input the exact cluster-condition groupings you wish to compare.

Expression Parameters that apply to all DEA subtabs:

- **Average Log2 Fold Change (log2FC) Threshold**
  Set the minimum log2 fold change required to consider a gene differentially expressed. *Note:* A log2FC value of |1| corresponds to a two-fold change in expression (i.e., FC = 2 or 0.5).
  Fold change (FC) is calculated as: FC = 2^log2FC Example: log2FC = 1 -> FC = 2, log2FC = -1 -> FC = 0.5
- **Minimum Percentage Threshold**
  Specify the minimum proportion of cells expressing the gene in either group (e.g., 25%).
- **Minimum Percentage Difference**
  Specify the minimum proportion difference of cells expressing the gene between the first and second group of comparison (e.g. 10% = 0.1).
- **Adjusted P-Value Threshold**
  Set the significance level for identifying differentially expressed genes, corrected for multiple testing.

Output that apply to all DEA subtabs

- **Differential Expression Table**
  A table listing all identified differentially expressed genes (DEGs) for each comparison.

- **Volcano Plots**
  One volcano plot per comparison, displaying the log2 fold change and adjusted p-values for each gene.
  - The plot title includes the number of DEGs detected.
  - The shape of each data point indicates whether the gene meets the **Minimum Percentage threshold**.
- **Violin Plot with Boxplot Overlay**
  A combined violin and boxplot visualizing the expression distribution of a selected gene across the groups involved in the comparison.

Enrichment Analysis (ORA)

This step performs **Over-Representation Analysis (ORA)** to identify biological pathways or functional categories that are significantly enriched in a set of **differentially expressed genes (DEGs)**. The analysis can be repeated multiple times using different parameters or gene sets.

1. **Select a DEA Result**
   Choose a previously generated **differential expression analysis (DEA)** result.
2. Any DEA result produced during the workflow can be selected, including those generated in the **Final Results** section.
3. **Choose a Database for Enrichment**
   Choose the biological database used to define gene sets for the enrichment analysis. Available databases include:
   - **Gene Ontology (GO): BP (Biological Process)**: biological pathways and cellular processes. **MF (Molecular Function)**: molecular activities of gene products. **CC (Cellular Component)**: cellular locations of gene products. **ALL**: combines all GO categories
   - **MSigDB Collections (C1–C8, H):** Curated gene-set collections containing pathways, regulatory signatures, and experimentally derived gene sets.
   - **KEGG:** Pathway based gene sets describing metabolic and signaling pathways.

<u>Note:</u> that gene sets originating from different databases may vary in their gene coverage and curation. As a result, enrichment results may differ depending on the selected database.

4. **Select a P-Value Adjustment Method**
   Choose a method to correct for multiple hypothesis testing:
   holm: Holm's correction (step-down Bonferroni)
   hochberg: Hochberg's correction (step-up Bonferroni)
   bonferroni: Very strict Bonferroni correction
   BH (default): Benjamini-Hochberg correction (commonly used; balances sensitivity and specificity)
   BY: Benjamini-Yekutieli correction (more conservative than BH)
   fdr: False Discovery Rate correction (equivalent to BH)
   none: No adjustment

The **Benjamini–Hochberg (BH)** method is commonly used because it balances sensitivity and specificity while controlling the false discovery rate.

5. **Set Adjusted P-Value Threshold**
   Specify the cutoff for statistical significance (e.g., **0.05**, which is commonly used).

The following results are generated for each ORA run:

Tables:

- **Enrichment summary tables:**
  Several tables summarize the enrichment results for different gene sets.
- **Pathway Table (Up- and Downregulated Genes)**
  Lists pathways significantly enriched in upregulated and downregulated genes. These are referred to as **Clusters**.
- **Table of Downregulated Genes per Pathway**
  Shows how many downregulated genes were found in each enriched pathway.
- **Table of Upregulated Genes per Pathway**
  Shows how many upregulated genes were found in each enriched pathway.
- **Table of All DEGs per Pathway**
  Shows pathways enriched using all DEGs, combining both up- and downregulated genes.

Plots:

- **Dot Plot**
  Displays the top 20 significant pathways where **y-axis** are the pathway names, **x-axis** are the user-selected variable (default: GeneRatio), **dot color** displays the adjusted p-value and **dot size** the number of genes found in the pathway.
- **Gene network (Gene Symbol Version)**
  Visualizes the top 3 most significant pathways and the genes they contain where **dot color (gene)** shows the expression value **and the dot size (pathway)** the number of associated genes.
- **Enrichment map (All DEGs)**
  Displays the top 20 significant pathways and their relationships based on shared genes. Where **dot color** is the adjusted p-value and **dot size** is the number of genes in each pathway
- **Expression-Specific Dot Plot**
  Dot plot showing the top 10 significant pathways enriched **within each expression** (up or downregulated DEGs).
- **Expression-Specific Gene network**
  Shows the top 3 significant pathways **per expression**, with gene symbols included.
- **Expression-Specific Enrichment map**
  Similar to the main enrichment map, but grouped **per expression**, visualizing pathway relationships and shared genes.

Gene Set Enrichment Analysis (GSEA)

This module performs **Gene Set Enrichment Analysis (GSEA)** to identify biological pathways that are significantly enriched at the top or bottom of a **ranked gene list** derived from differential expression results. Unlike Over-Representation Analysis (ORA), which uses a predefined list of significant genes, GSEA considers the **entire ranked gene list**, allowing the detection of subtle but coordinated changes in gene expression. The analysis can be repeated multiple times using different parameters or gene set sources.

- **Select a DEA Result**
  Choose a previously generated **differential expression analysis (DEA)** result. Any

DEA result produced during the workflow can be used, including those generated in the **Final Results** section.

- **Choose a Category for Analysis:** Choose the gene set collection used for the enrichment analysis. Two options are available:
    - **MSigDB Gene Sets:** Curated gene-set collections from the **Molecular Signatures Database (MSigDB)**. These collections include pathways, regulatory targets, biological processes, and hallmark gene signatures. Available categories include:
      C1-C8: curated gene-set collections covering biological processes, regulatory signatures, and pathway databases
      H (Hallmark): concise gene sets representing well-defined biological states or processes. More information on the available categories (MSigDB)
    - **Custom Gene Sets**
      Users may upload their own gene sets in **Excel (.xlsx) format**. The file should be structured such that: each column represents a gene set (e.g., a pathway, module, or signature) and each row contains genes belonging to that gene set. *Example:*

B Cells	T Cells
ENSMUSG00000026395	**ENSMUSG00000026395**
ENSMUSG00000030724	**ENSMUSG00000032093**
ENSMUSG00000003379	**ENSMUSG00000032094**

- **Gene identifier type:**
  The gene identifiers used in the uploaded gene-set file must match the identifier format used in the dataset.Supported formats include: **Ensembl IDs and Gene symbols.** Ensembl IDs are generally recommended because they are stable identifiers and reduce ambiguity caused by gene symbol synonyms. If the Seurat object was generated during **Phase 1 of PICASSA**, gene identifiers are typically expected to be **Ensembl IDs**. If an external Seurat object is used instead, the identifiers in the uploaded gene-set file must match the gene identifiers present in that dataset. Gene identifiers can be converted to **Ensembl IDs** using the **Gene Conversion** tool available in PICASSA.
- **Set Adjusted P-Value Threshold**
  Define the significance cutoff (commonly set at **0.05**).
- **Choose a Ranking Metric**
  Select how genes should be ranked for GSEA:
    - **Average Log2FC**: Commonly used and stable
    - **Rank Metric**: Uses a combination of log2FC and adjusted p-values, but may fail in some cases

The following results are generated for each GSEA run:
Tables:

- **Enriched Pathway Table**
  Lists the pathways that are significantly enriched at either end of the ranked gene list.
- **Gene Count Table**
  Shows how many genes from the comparison were found in each enriched pathway.
- **Barplot**
  A barplot visualizing the number of genes found in each enriched pathway (based on the gene count table above).

Plots:
All plots allow selection of significant pathways via dropdown menus or filtering options.

- **Dot Plot**
  Displays the top 20 significant pathways where **y-axis** are the pathway names **x-axis** is the user-selected variable (default: NES score), **dot color** displays the adjusted p-value and **dot size** the number of genes in the pathway
- **Enrichment map**
  Visualizes relationships between the top 20 significant pathways based on gene overlap where **dot color** is the adjusted p-value and **dot size** is the number of genes in each pathway
- **Gene network (Gene Symbol View)**
  Shows the top 3 most significant pathways with associated genes where **dot color (gene)** shows the expression value and **dot size (pathway)** displays the number of genes in the pathway
- **GSEA Plot**
  Visualizes the **Running Enrichment Score** (y-axis) across the **Rank in Ordered Dataset** (x-axis). It also shows the **Ranked List Metric** below, to view enrichment for multiple pathways to determine whether they are enriched at the top or bottom of the ranked gene list

Drug-Gene Interaction Analysis

This analysis identifies potential drug-gene interactions using DEA results and the **Drug-Gene Interaction Database (DGIdb)**.

1. **Select a DEA Result**
   Choose any differential expression analysis result previously generated, including those from the *Final UMAPs* section.

The **Drug-Gene Interaction Database (DGIdb)** is a comprehensive resource that curates known drug-gene interactions and potential druggable targets. It integrates data from multiple sources, including peer-reviewed literature, public databases, and expert curation. This makes it a powerful tool for supporting drug discovery and precision medicine.
Explore the DGIdb
This panel produces a table and a network plot.

- **Drug-Gene Interaction Table**
  Displays the interactions found between selected genes and known drugs.
- **Network Plot**
  A visual representation of the drug-gene interaction network where **red nodes** represent drug names **blue nodes** represent gene names and **edge width** corresponds to the interaction score between the drug and gene

Trajectory Analysis (Monocle3)

Trajectory analysis allows the study of cell lineage progression by ordering cells along a pseudotime axis based on their transcriptomic similarity. This approach helps identify differentiation pathways, lineage branch points and dynamic gene expression changes during cellular transitions. The trajectory is inferred using Monocle3, which constructs a trajectory graph in a low-dimensional embedding (UMAP) and assigns a pseudotime value to each cell.

1. **Select Clusters for Analysis**
   Users may optionally select specific clusters to include in the trajectory analysis, otherwise, all clusters in the dataset are used. Restricting the analysis to selected clusters can help focus on a particular lineage or biological process.
2. **Select the Initial Cluster**
   The initial cluster defines the starting point of the trajectory (root of pseudotime) and should be selected based on prior biological knowledge, as cells from this cluster are used to initialize pseudotime ordering.
3. **Run Trajectory by Condition**
   Trajectory analysis can optionally be performed separately for each experimental condition. This option is useful for comparing trajectory structures between groups.
4. **Identify Genes Varying Along the Trajectory**
   Detect genes whose expression changes along the trajectory. This is done using the Geary's C spatial autocorrelation test, which measures how similar gene expression is among neighboring cells in the trajectory graph. Genes with strong spatial structure along the trajectory are considered trajectory-associated genes.

Trajectory analysis generates multiple outputs that are designed to characterize cell state transitions, lineage relationships, and gene expression dynamics along pseudotime:

Trajectory Visualization

Three types of nodes appear in the trajectory graph:

- **White circles**: Root nodes (starting points of pseudotime)
- **Black circles**: Branch points where differentiation paths diverge (lineage split points)
- **Light gray circles**: Leaf nodes representing terminal states (endpoints of differentiation)

The following plots are generated:

- **Cluster Trajectory Plot:** A UMAP showing the selected clusters together with the inferred trajectory graph connecting related cell states.
- **Pseudotime Trajectory Plot:** Cells are colored according to their pseudotime values, showing how cells progress along the trajectory.
- **Pseudotime with Branch Nodes:** A trajectory plot highlighting branch points, roots, and terminal states, providing insight into lineage branching events.
- **Pseudotime Density per Cluster:** A density plot showing the distribution of pseudotime values across clusters where **x-axis** shows the pseudotime and **y-axis** the cluster identities.

Top 5 Trajectory Genes

If "Find genes that are differentially expressed across a single-cell trajectory?" is"Yes", then the top five genes whose expression changes along the trajectory are displayed, along with a results table for all tested genes. Each gene is evaluated using the Geary's C spatial

autocorrelation statistic, which measures the similarity of gene expression among neighboring cells along the trajectory. Lower Geary's C values indicate stronger spatial structure in gene expression, whereas higher values suggest weaker spatial patterns. Genes are ranked based on their statistical significance (q value) and their Geary's C score, and the five most significant genes are visualized.

Feature Plots of Top 5 Genes
Five feature plots display the expression of the top trajectory associated genes across the UMAP embedding. These plots help identify where along the trajectory each gene is expressed.

Combined Expression Plot
A single UMAP displays the expression of all five genes simultaneously. Expression values are scaled as %Max, meaning each gene's expression is normalized relative to its maximum observed value. This allows easier comparison of expression patterns across genes.

Gene Statistics Table
Each row in the table represents one gene tested along the trajectory. The **status** column indicates whether the statistical test was successfully executed. The p_value column reports the p-value obtained from the Geary's C spatial autocorrelation test, which evaluates whether gene expression shows significant spatial correlation among neighboring cells along the trajectory. The geary_test_statistic column shows the statistic used to compute the p-value, while geary_C represents the Geary's C spatial autocorrelation score, where lower values indicate stronger spatial structure in gene expression. The gene column contains the gene identifier from the seurat object; if both gene symbol and Ensembl ID are available, additional columns display each identifier separately. The q_value column provides the adjusted p-value calculated using the Benjamini–Hochberg false discovery rate (FDR) correction and is used to rank genes by statistical significance. Genes with low q-values and low Geary's C scores exhibit the strongest trajectory-associated expression patterns.

Visualizing Specific Genes
Users can also enter custom gene names to visualize their expression dynamics along pseudotime. This allows tracking of genes of interest across the trajectory and identifying where they become activated or repressed during differentiation.

LIANA: Ligand-Receptor Interaction Analysis

The LIANA tool enables the inference of cell-cell communication by integrating results from multiple ligand-receptor (LR) analysis methods. It provides robust, consensus-based interaction scores between cell types.

1. **Annotation Source Selection**
   Choose which annotation column to use for identifying cell types in the analysis:
   - **final_clusters**: Numeric cluster Ids
   - **scType**: Annotations from the ScType tool
   - **scMayo**: Annotations from scMayoMap
2. **Select a Resource**
   Choose the database or knowledge resource to define ligand-receptor pairs.

3. **Select Methods**
   Choose which inference methods to use.
   - **natmi** and **sca** are selected by default and are recommended for balanced performance.

More information about LIANA Results

LIANA panel generates a table and multiple interactive plots such as heatmap, dot plot and a chord diagram:

- **Summary Table**
  - Displays the **top 10 ligand-receptor interactions** based on multiple inference methods.
  - Each row includes individual method scores and an **aggregate consensus rank** (aggregate_rank), calculated using the RobustRankAggreg (RRA) method.
  - The **RRA score** can be interpreted like a p-value: interactions consistently ranked high across methods receive lower scores (higher confidence).
- **Heatmap**
  - Shows the **frequency of interactions** between each pair of potential communicating cell types.
  - Interactions are filtered by the aggregate_rank, reflecting their significance and consistency.
  - This is a useful overview to identify which cell types communicate most frequently and robustly.
- **Dot Plot**
  - Visualizes the **top 20 ligand-receptor interactions** between source and target cell types.
  - Two key metrics are displayed:
  - **Specificity**: Indicates how uniquely an interaction occurs between a specific cell pair (e.g., NATMI's specificity, CellPhoneDB's p-values).
  - **Magnitude**: Reflects the expression level of the ligand and receptor pair (e.g., SingleCellSignalR's LRscore ranging from 0 to 1).

Note: Interactions not expressed in at least 10% of cells in both clusters are excluded and shown as missing dots.

- **Chord Diagram**
  - A circular plot that visually represents the frequency of interactions between cell types, similar to the heatmap.
  - Useful for understanding directional communication (source → target) at a glance.

Report

The report can be downloaded at any time from the movable panel located in the top-right corner, allowing all performed steps to be saved.